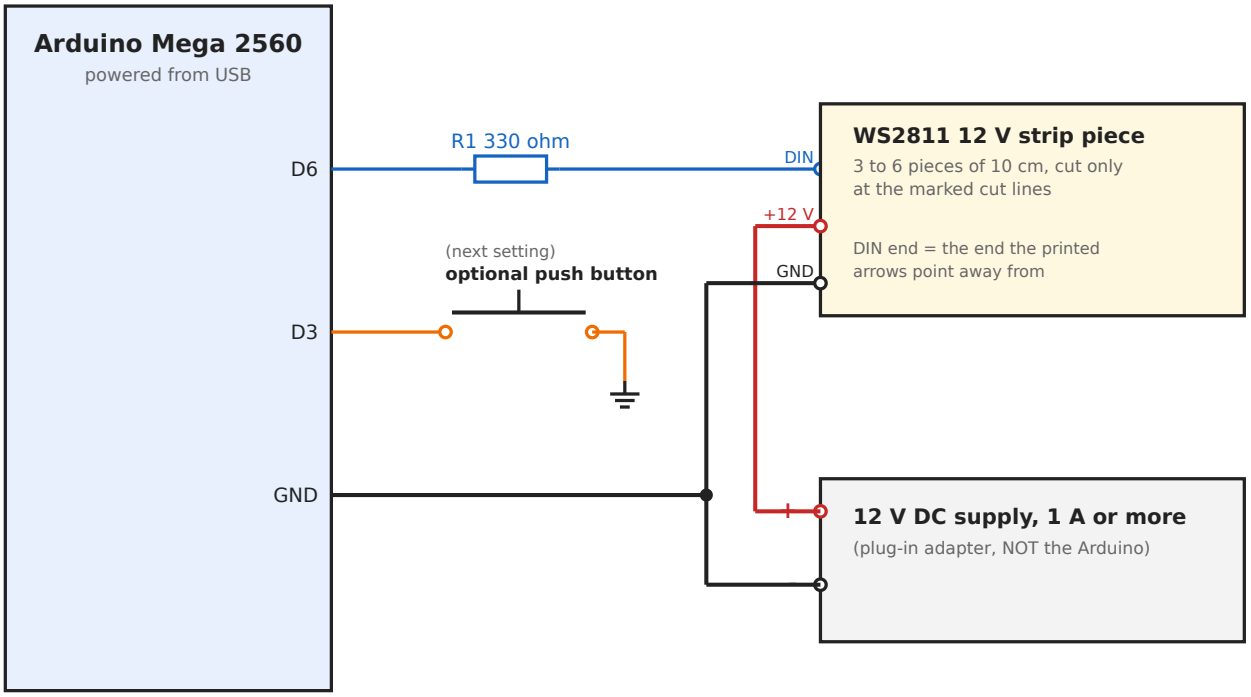
LED strip test - wiring (Arduino Mega 2560 + a short piece of the 12 V strip)



Connections

1. Arduino D6 -> 330 ohm resistor -> DIN of the strip piece (the end where the printed arrows point away from).
2. The strip's +12 V comes from the 12 V supply, NEVER from the Arduino 5V pin. Do not connect 12 V to any Arduino pin.
3. GND of the 12 V supply, GND of the strip and GND of the Arduino are connected together (common ground).
4. The Arduino is powered from USB. The push button (D3 to GND) is optional: it moves to the next setting.
5. Optional: a 1000 uF capacitor rated 25 V or more across the strip's +12 V and GND (plus to +12 V, stripe = minus).
6. The built-in LED (pin 13) blinks the setting number 1..12 at the start of each setting; the Serial Monitor (115200) prints it too. Tell which setting numbers show red, green, blue in order and how many LEDs light up in each step.
7. If nothing lights: check the strip direction (DIN end), the common GND and that the 12 V supply is on.